EIN 4343 Inventory and Supply Chain Systems

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The Role of Logistics

- Process of planning, implementing, and controlling the efficient, cost-effective flow and storage of materials, in-process inventory, finished goods, and related information from point of origin to point of consumption

- Need to consider inventory and transportation decisions in an integrated manner

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Logistics decisions

- Transportation mode selection
- Shipment size specification
- Carrier selection

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

LOGISTICS: TRANSPORTATION MODE SELECTION

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Logistics decisions

- Transportation mode selection
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- Carrier selection

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Inventory holding cost

- There is not an “inventory holding cost formula.”
- The structure of the inventory holding cost must be discerned from the problem statement.

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Transportation cost

- The transportation cost may be charged
  - Per shipment, i.e., a type of fixed cost.
  - Per unit, i.e., a type of variable cost.
  - etc.

- Hence, there is not a “transportation cost formula.”

- The structure of transportation cost must be discerned from the problem statement.

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In-transit inventory holding cost

The inventory holding cost incurred while items are in transit in the logistics pipeline is known as in-transit inventory holding cost and is given by

$$h\lambda\tau$$

where

h: the <u>annual</u> in-transit inventory holding cost,

λ: the annual demand, and

τ: lead time in <u>years</u>.

Note that the terms need to be time consistent, i.e., expressed in annual terms!

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Togo Shoe Co.

Togo Shoe Co. produces a line of men's shoes. The company owns a warehouse in San Francisco that supplies the retailers on the West Coast. The warehouse operates for 360 days/year. Togo has a policy of keeping 1,000 pairs of shoes as safety stock in the San Francisco warehouse for every day of transportation.

A pair of shoes has an average value of $30 and inventory carrying costs are calculated using an interest rate of 30% per year in California and 12% per year while in transit. 70,000 pairs of shoes are sold per year out of the warehouse on the West Coast.

The company wishes to select the mode of transportation that will minimize the total logistics costs associated with shipping the shoes from the manufacturing facility to the warehouse and storing the shoes at the warehouse. The company is considering the following transportation modes:

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Togo Shoe Co.

Mode	Transportation Rate ($/unit)	Door-to-Door Transit Time (days)	Number of Shipments per Year
Rail	0.10	21	10
Intermodal	0.15	14	20
Truck	0.75	5	20
Air	1.60	2	40

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Togo Shoe Co.: Rail

Shipment Quantity:

$$q = \frac{70{,}000}{10} = 7{,}000 \text{ units}$$

Safety Stock Amount:

$$s = (21)(1{,}000) = 21{,}000\text{units}$$

Inventory Holding Cost In-transit:

$$(0.12)(30)(70{,}000)\left(\frac{21}{360}\right) = \$14{,}700$$

Transportation Cost:

$$(0.10)(70{,}000) = \$7{,}000$$

Inventory Holding Cost in California:

$$(0.30)(30)\left(21{,}000 + \frac{7{,}000}{2}\right) = \$220{,}500$$

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Togo Shoe Co.: Piggyback

Shipment Quantity:

$$q = \frac{70{,}000}{20} = 3{,}500 \text{ units}$$

Safety Stock Amount:

$$s = (14)(1{,}000) = 14{,}000 \text{ units}$$

Inventory Holding Cost In-transit:

$$(0.12)(30)(70{,}000)\left(\frac{14}{360}\right) = \$9{,}800$$

Transportation Cost:

$$(0.15)(70{,}000) = 10{,}500$$

Inventory Holding Cost in California:

$$(0.30)(30)\left(14{,}000 + \frac{3{,}500}{2}\right) = \$141{,}750$$

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Togo Shoe Co.: Comparison

Mode	q	s	In-transit Inv Hold	Transp	Inv Hold @ CA	Total Cost
Rail	7,000	21,000	$14,700	$7,000	$220,500	$242,200
Piggyback	3,500	14,000	$9,800	$10,500	$141,750	$162,050
Truck	3,500	5,000	$3,500	$52,500	$60,750	$116,750
Air	1,750	2,000	$1,400	$112,000	$24,875	$139,275

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LOGISTICS: SHIPMENT SIZE SELECTION

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Logistics decisions

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Caterpillar

Caterpillar uses a 3PL provider for the shipment of engines to Caterpillar's warehouse in Illinois. Currently, the shipments of 20,000 lbs or more are moved at a rate of $5/cwt. If the shipment size falls below 20,000 lbs, a rate of $9/cwt applies.

The distribution planning manager of Caterpillar has received a rate reduction offer from their 3PL. If Caterpillar agrees to move 40,000 lbs or more in each shipment, the rate will be $3/cwt.

For the inventory of engines on hand at the warehouse, two types of inventory costs are incurred:

- the opportunity cost of keeping stock and
- out-of-pocket expenses to handle the stock.

Should Caterpillar adjust the replenishment quantity so that the new rate can be used?

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Caterpillar

Information item	Parameter value
Annual demand	5000 engines/year
Warehouse replenishment orders	43 orders/year
Weight of each engine with crate	175 lbs/engine
Standard cost of engine in the warehouse	$200/engine
Stock replenishment order costs	$15/order
Inventory carrying rate	25%/year
Out-of-pocket inventory holding cost	$0.30/cwt
Warehouse space	Unlimited

cwt: per cent weight is an approach to determine transportation cost such that the unit cost incurred/charged for every 100 lbs shipped.

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Caterpillar: Current

Shipment Quantity and Weight:

$$q_c = \frac{5{,}000}{43} = 116.28 \text{ engines}$$

$$w_c = (116.28)(175) = 20{,}343.84 \text{ lbs}$$

Ordering Cost:

$$(15)\left(\frac{5{,}000}{116.28}\right) = \$645$$

Transportation Cost:

$$(5)\left(\frac{(116.28)(175)}{100}\right)\left(\frac{5{,}000}{116.28}\right) = \$43{,}750$$

Inventory Holding Cost:

$$(0.25)(200)\left(\frac{116.28}{2}\right) = \$2{,}907$$

Inventory Handling Cost:

$$(0.30)\left(\frac{(116.28)(175)}{100}\right)\left(\frac{5{,}000}{116.28}\right) = \$2{,}625$$

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Caterpillar: Proposed

Shipment Quantity and Weight:

$$q_p = \frac{40{,}000}{175} = 228.57 \text{ engines}$$

$$w_p = 40{,}000 \text{ lbs}$$

Ordering Cost:

$$(15)\left(\frac{5{,}000}{228.57}\right) = \$328$$

Transportation Cost:

$$(3)\left(\frac{40{,}000}{100}\right)\left(\frac{(5{,}000)}{228.57}\right) = \$26{,}250$$

Inventory Holding Cost:

$$(0.25)(200)\left(\frac{228.57}{2}\right) = \$5{,}714$$

Inventory Handling Cost:

$$(0.30)\left(\frac{(228.57)(175)}{100}\right)\left(\frac{(5{,}000)}{228.57}\right) = \$2{,}625$$

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Caterpillar: Comparison

Setting	q units (w lbs)	Order Cost	Transp Cost	Inv Carry Cost	Inv Hand Cost	Total Cost
Current	116.28 (20,343.84)	$645	$43,750	$2,907	2,625	$49,927
Proposed	228.57 (40,000)	$328	$26,250	$5,714	2,625	$34,917

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Janus: God of duality

- Of beginnings and endings
- Of the past and the future
- Ability to see “both” sides of an issue…

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Kroeger and Walmart

- Digital price tags by Kroeger and Walmart

https://www.nytimes.com/2024/10/23/business/kroger-walmart-facial-recognition-prices.html?searchResultPosition=1

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Kroeger and Walmart

- Body scans, face scans
  - Both Walmart and Kroeger have experimented with them in the past
  - Not currently active

- What could happen if the digital price tags were to change depending on the customer?

https://www.nytimes.com/2024/10/23/business/kroger-walmart-facial-recognition-prices.html?searchResultPosition=1

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Kroeger

- Digital price tags can be used
  - to offer discounts,
  - To place items on sales if they are reaching their expiration date

- Dynamic pricing
  - Changes in price as a function of changes in the supply and/or demand processes

https://www.nytimes.com/2024/10/23/business/kroger-walmart-facial-recognition-prices.html?searchResultPosition=1

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LOGISTICS: CARRIER SELECTION

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Transportation management decisions

- Transportation mode selection
- Shipment size specification
- Carrier selection

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Mavi Jeans

Mavi Jeans is a producer of jeans for men and women. Mavi has a distribution center in New York that receives shipments from the production facilities in Bursa, Turkey. From there, the jeans are routed to department stores and Mavi stores in North America. The demand in the North American market is approximately 96,000 pairs per year. The facility operates 48 weeks throughout the year. The cost of placing an order with the manufacturing facility in Turkey is about $150 per order. The cost of a pair of jeans is $20 per unit.

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Mavi Jeans

The distribution center manager in New York, Dominique Blue, estimates that the company will incur an inventory holding cost at a rate of 35 percent a year while the jeans are in the distribution center. Ms. Blue plans to use the EOQ to replenish the inventory and a reorder-point type inventory policy to place the replenishment orders. She states that the stock-out costs are unknown since the company has recently entered the North American market. However, following the practice of the company in the European market, she believes that the business will strive toward a Type I service level of 90 percent in the North American market as well. Historically, the demand during lead time for Mavi is approximately normally distributed.

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Mavi Jeans

Ms. Blue has conducted a survey of the carriers that provide transatlantic services and has narrowed down her choice to two companies that can make deliveries from the manufacturing plant to the distribution center. Horoz Logistics is more expensive, but its delivery performance has been rated the highest in the industry, whereas Speedy Carriers is less reliable. The following information is available:

	Horoz Logistics	Speedy Carriers
Average delivery time (weeks)	6	6
Standard deviation of delivery time (weeks)	1	4
Rate ($/cwt)	25.00	12.50

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Mavi Jeans

Horoz Logistics offers unlimited liability for potential damage to the jeans. As a result, Mavi will incur an in-transit inventory carrying rate of 8 percent per year.

Speedy Carriers offers limited liability, thus Mavi will incur an in-transit inventory carrying rate of 12 percent per year.

Which carrier should Mavi choose?

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Mavi Jeans: Horoz Logistics

Shipment Quantity, DDLT, and safety stock:

$$q_{HL} = \sqrt{\frac{(2)(150)(96{,}000)}{(.35)(20)}} = 2{,}028.37 \text{ pairs}$$

$$\mu_L^{HL} = (6)(2{,}000) = 12{,}000 \text{ pairs}$$

$$\sigma_L^{HL} = \sqrt{(6)(0^2) + (2{,}000^2)(1^2)} = 2{,}000 \text{ pairs}$$

$$\alpha = F(r) = 0.90 \quad \Rightarrow z = 1.2816$$

$$s^{HL} = (2{,}000)(1.2816) = 2{,}563.20 \text{ pairs}$$

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Mavi Jeans: Horoz Logistics

Ordering Cost:

$$(150)\left(\frac{96{,}000}{2{,}028.37}\right) = \$7{,}099$$

Transportation Cost:

$$(25)\left(\frac{(96{,}000)(4)}{100}\right) = \$96{,}000$$

Inventory Holding Cost:

$$(0.35)(20)\left(2{,}563.20 + \frac{2{,}028.37}{2}\right) = \$25{,}042$$

In-transit Inventory Holding Cost:

$$(0.08)(20)(96{,}000)\left(\frac{6}{48}\right) = \$19{,}200$$

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Mavi Jeans: Speedy Carriers

Shipment Quantity, DDLT, and safety stock:

$$q_{SC} = \sqrt{\frac{(20)(150)(96{,}000)}{(.35)(20)}} = 2{,}028.37 \text{ pairs}$$

$$\mu_L^{SC} = (6)(2{,}000) = 12{,}000 \text{ pairs}$$

$$\sigma_L^{SC} = \sqrt{(6)(0^2) + (2{,}000^2)(4^2)} = 8{,}000 \text{ pairs}$$

$$\alpha = F(r) = 0.90 \quad \Rightarrow z = 1.2816$$

$$s^{SC} = (8{,}000)(1.2816) = 10{,}252.80 \text{ pairs}$$

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Mavi Jeans: Speedy Carriers

Ordering Cost:

$$(150)\left(\frac{96{,}000}{2{,}028.37}\right) = \$7{,}099$$

Transportation Cost:

$$(12.5)\left(\frac{(96{,}000)(4)}{100}\right) = \$48{,}000$$

Inventory Holding Cost:

$$(0.35)(20)\left(10{,}252.80 + \frac{2{,}028.37}{2}\right) = \$78{,}869$$

In-transit Inventory Holding Cost:

$$(0.12)(20)(96{,}000)\left(\frac{6}{48}\right) = \$28{,}800$$

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Mavi Jeans: Comparison

Carrier	q units	s units
Horoz Log	2028.37	2,563.20
Speedy Car	2028.37	10,252.80

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Mavi Jeans: Comparison

Carrier	q units	s units
Horoz Log	2028.37	2,563.20
Speedy Car	2028.37	10,252.80

Carrier	Order	Transp	Inv Hold	In-transit Inv Hold	Total Cost
Horoz Log	$7,099	$96,000	$25,042	$19,200	$147,341
Speedy Car	$7,099	$48,000	$78,869	$28,800	$162,768

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